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Inventor(s)	Kadobe; Masato et al.

Piping structure and processing apparatus

Abstract

A piping structure includes a plurality of pipes that connects a plurality of process modules disposed adjacent to each other at a first room and a plurality of vacuum pumps disposed at a second room below the first room to be corresponding to the plurality of process modules, respectively. The plurality of pipes are divided into a plurality of blocks in a height direction, and the plurality of pipes used for blocks in an identical height have an identical shape.

Inventors:	Kadobe; Masato (Iwate, JP), Nitadori; Hiromi (Iwate, JP)
Applicant:	Tokyo Electron Limited (Tokyo, JP)
Family ID:	1000008778949
Assignee:	TOKYO ELECTRON LIMITED (Tokyo, JP)
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Primary Examiner: Anderson, II; Steven S

Attorney, Agent or Firm: Venjuris, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority from Japanese Patent Application No. 2021-052327 filed on Mar. 25, 2021 with the Japan Patent Office, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a piping structure and a processing apparatus.

BACKGROUND

(3) In the related art, a processing apparatus has been known in which a processing unit that includes a plurality of process modules disposed adjacent to a first room and a loader module accommodating a carrier that is disposed in the first room and accommodates substrates to be processed by the plurality of process modules, and a plurality of pump units that is disposed corresponding to the plurality of process modules in a second room adjacent to the first room, respectively, are provided, and the length of a connecting pipe for connecting the corresponding plurality of process modules and the plurality of pump units is the same (see, e.g., Japanese Patent Laid-Open Publication No. 2020-113629).

SUMMARY

(4) In order to achieve the above-described objective, a piping structure according to an embodiment of the present disclosure includes a plurality of pipes that connects a plurality of process modules disposed adjacent to each other at a first room and a plurality of vacuum pumps disposed at a second room below the first room to be corresponding to the plurality of process modules, respectively. The plurality of pipes are divided into a plurality of blocks in a height direction, and the plurality of pipes used for blocks in an identical height have an identical shape.

(5) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a view illustrating an example of the disposition of a processing apparatus according to a first embodiment of the present disclosure.
- (2) FIG. 2 is a perspective view illustrating an example of a piping structure according to the present embodiment.
- (3) FIG. 3 is a front view of the piping structure according to the present embodiment.
- (4) FIG. 4 is a side view of the piping structure according to the present embodiment.
- (5) FIG. 5 is an exploded view illustrating one pipe of the piping structure according to the embodiment of the present disclosure in an exploded manner.
- (6) FIG. 6 is a view illustrating an example of the configuration of a flange.
- (7) FIG. 7 is a view illustrating an example of partial pipes of a piping structure according to a second embodiment of the present disclosure.
- (8) FIG. 8 is a view illustrating an example of an opening of a clean room facility structure.
- (9) FIGS. 9A and 9B are views illustrating a swing width in a pipe.
- (10) FIG. 10 is a view illustrating a concept of route setting.
- (11) FIGS. 11A and 11B are views illustrating an example of the shape of a pipe in which a fluctuation range is set to be small.
- (12) FIGS. 12A and 12B are views illustrating an example of the shape of a pipe in which a fluctuation range is set to be large.

DETAILED DESCRIPTION

- (13) In the following detailed description, reference is made to the accompanying drawings, which form a part thereof. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made without departing from the spirit or scope of the subject matter presented here.
- (14) Hereinafter, embodiments for implementing the present disclosure will be described with reference to the accompanying drawings.

First Embodiment

- (15) FIG. 1 is a view illustrating an example of the disposition of a processing apparatus according to a first embodiment of the present disclosure. The processing apparatus according to the present embodiment includes a process module 40, a vacuum pump 50, and a piping structure 60. The process module 40 is installed in the first room 10, and the vacuum pump 50 is installed in the second room 20 below the first room 10. A clean room facility structure 30 is provided between the first room 10 and the second room 20.
- (16) The process module 40 is a module for processing a substrate (not illustrated), and performs a process for processing such as a film formation, an annealing, or an etching, or a process for

manufacturing on the substrate. A plurality of process modules **40** is provided in parallel to improve a processing efficiency. In FIG. **1**, four process modules **40** are provided. In addition to the process modules **40**, a loader **41** is provided to load the substrate into each process module **40**. The process modules **40** may process various substrates depending on the usage, and descriptions will be made on an example of using a silicon wafer (hereinafter, referred to as a “wafer”) as a substrate.

(17) A module that performs various processes may be used as the process module **40**, but in the present embodiment, descriptions will be made on the case where a vertical heat treatment apparatus is used as the process module. The process module **40** accommodates a substrate holder capable of loading a plurality of substrates. For example, each process module **40** is configured as a module capable of processing **50** to **200** substrates at a time. By disposing a plurality of such process modules **40** in parallel and executing processes at the same time, the number of processed substrates may be increased. Therefore, it is possible to dispose a plurality of process modules **40** adjacent to each other and perform a substrate process in parallel.

(18) The loader **41** is a device for receiving and storing a plurality of FOUPs, and the productivity per unit area may be improved by associating one loader **41** with a plurality of process modules **40**. Therefore, it is required that the plurality of process modules **40** are disposed adjacent to each other to have a small footprint as a whole as much as possible.

(19) Since the process module **40** is a module for performing a vacuum process, it is required to connect a vacuum pump **50** to each process module **40** and evacuate the inside of the process module **40**. Therefore, a one-to-one vacuum pump **50** is provided corresponding to each process module **40**, and four vacuum pumps **50** are installed corresponding to four process modules **40**.

(20) The vacuum pump **50** is an exhaust unit for evacuating the processing chamber of the process module **40**, and a vacuum pump **50** such as a dry pump, a turbo molecular pump, or a cryopump may be used according to a process.

(21) As illustrated in FIG. **1**, in many cases, the process module **40** may be installed in the first room **10**, and the vacuum pump **50** may be installed in the second room **20** downstairs. In this case, pipes **61** to **64** are installed corresponding to the respective pairs of the process module **40** and the vacuum pump **50**, and each process module **40** and each vacuum pump **50** are connected to each other. Therefore, in FIG. **1**, four pipes **61** to **64** are provided corresponding to the process modules **40** and the vacuum pumps **50**.

(22) Here, when the process module **40** in the first room **10** and the vacuum pump **50** in the second room **20** downstairs are connected by a pipe **60**, a clean room facility structure **30** may be provided between the first room **10** and the second room **20**.

(23) The clean room facility structure **30** is provided with openings **31**, and it is necessary to install pipes **61** to **64** through the openings **31**. That is, it is necessary to connect the process module **40** and the vacuum pump **50** with each other under the constraint condition of the position of the opening **31** of the clean room facility structure **30**.

(24) In FIG. **1**, an example is given in which the vacuum pumps **50** are disposed adjacent to each other directly under the process module **40**, but there may be a case where the vacuum pumps **50** are not necessarily disposed together due to restrictions such as columns and walls. In such a case, when the pipe easiest to use is selected and connected based on a disposition relationship between the process module **40** and the vacuum pump **50**, the shape and length of the pipe differ for each process module **40**, and due to the change in exhaust conditions, a situation may occur in which the uniformity of the film forming process and the etching process performed in each process module may not be maintained.

(25) Therefore, in the piping structure according to the present embodiment, a configuration is adopted in which the shapes and lengths of the plurality of pipes **61** to **64** are unified so that the conductance in the plurality of pipes **61** to **64** is the same. Hereinafter, this point will be described in detail.

(26) FIG. 2 is a perspective view illustrating an example of the piping structure **60** according to the present embodiment. As illustrated in FIG. 2, the piping structure **60** includes four pipes **61** to **64**, but the directions of the inclined portions of the pipes **61** to **64** are different from each other.

(27) FIG. 3 is a front view of the piping structure according to the present embodiment. As illustrated in FIG. 3, the piping structure **60** includes pipes **61** to **64**, each of which is divided into three regions A to C in the height direction. The regions A to C are divided pipes of the pipes **61** to **64**, respectively, and are cut portions connected by flanges. That is, the regions A to C are units that constitute a pipe as one component.

(28) Focusing on the region A, the four pipes **61** to **64** have divided pipes **611** to **641**, respectively. The divided pipes **611** to **641** are disposed in different directions, but all have the same shape. More specifically, the partial pipes **611** to **641** have inclined portions **611a** to **641a** and straight portions **611b** to **641b**, respectively, but the inclined portions **611a** to **641a** all have the same shape and the straight portions **611b** to **641b** also have the same shape.

(29) Specifically, the inclined portions **611a** to **641a** are different in the inclination directions, the inclined portion **611a** is slightly inclined to the right, the inclined portion **621a** is slightly inclined to the left, the inclined portion **631a** is inclined to the left at a slightly larger angle than the inclined portion **621a**, and the inclined portion **641a** is inclined to the front.

(30) Meanwhile, the straight portions **611b** to **641b** all extend in the vertical direction and are differently disposed, but have the same shape.

(31) Therefore, although the directions and positions of the partial pipes **611** to **641** are different from each other, the shapes thereof are all the same. When the partial pipes **611** to **641** have the same shape, since the conditions as a gas flow path are the same, the conductance thereof is the same.

(32) Similarly, in the region B, the pipes **61** to **64** have partial pipes **612** to **642**, respectively. The partial pipes **612** to **642** have inclined portions **612a** to **642a** and straight portions **612b** to **642b**, respectively. Also in the region B, the inclination directions of the inclined portions **612a** to **642a** are different from each other, the inclined portion **612a** faces the left side, and the inclined portion **622a** faces slightly to the left side. The inclined portion **632a** faces the front, and the inclined portion **642a** faces the left side at an angle smaller than the inclined portion **612a** and larger than the inclined portion **622a**. However, since the shapes of the inclined portions **612a** to **642a** are all the same, the conductance of the gas at the time of exhaust is the same.

(33) Although the straight portions **612b** to **642b** are different in the disposed positions, the straight portions **612b** to **642b** all extend in the vertical direction and have the same length and thickness. Therefore, the conductance of the partial pipes **612** to **642** is the same.

(34) Since the shapes of the inclined portions **612a** to **642a** are all the same and the shapes of the straight portions **612b** to **642b** are all the same, the conductance of the partial pipes **612** to **642** is also the same.

(35) Regarding the region C, the partial pipes **613** to **643** have straight portions **613a** to **643a**, inclined portions **613b** to **643b**, and straight portions **613c** to **643c**, respectively. The partial pipes **611** to **641** and the partial pipes **612** to **642** in the regions A and B are respectively combinations of the inclined portions **611a** to **641a** and **612a** to **642a**, and the straight portions **611b** to **641b** and **612b** to **642b**, that is, a combination of one inclined portion and one straight portion. However, the partial pipes **613** to **643** are different from the partial pipes **611** to **641** and the partial pipes **612** to **642** in the regions A and B in that the partial pipes **613** to **643** are combinations of two straight portions **613a** to **643a** and **613c** to **643c** and one inclined portion **613b** to **643b**.

(36) As described above, the combinations of the straight portions **613a** to **643a** and **613c** to **643c** and the inclined portions **613b** to **643b** in the partial pipes **613** to **643** may be variously prepared depending on the usage.

(37) Also in the partial pipes **613** to **643**, the shapes of the straight portions **613a** to **643a** are the same, the shapes of the inclined portions **613b** to **643b** are also the same, and the shapes of the

straight portions **613c** to **633c** are also the same. Thus, the conductance of the partial pipes **613** to **641** is also the same.

(38) Further, although the relative positions of the straight portions **613a** to **643a** and the inclination directions of the inclined portions **613b** to **643b** are different from each other, the positions of the straight portions **613c** to **643d** connected to the vacuum pump **50** with respect to the vacuum pump **50** are the same.

(39) The piping structure **60** according to the present embodiment may start from the connection position of the process module **30** and may be set to a position suitable for the disposition of the vacuum pump **50** at the final arrival of the vacuum pump **50**, and the intermediate routes may be disposed differently. Then, while passing through such different intermediate routes, the shapes in all the regions A to C may be made the same, and the conductance therein may be kept constant.

(40) As described above, in all the regions A, B, and C in the height direction, the partial pipes **611** to **641**, the partial pipes **612** to **642**, and the partial pipes **613** to **643**, respectively, have the same shape, so that the conductance of these partial pipes is the same. Therefore, the conductance of the pipes **61** to **64** is also the same. In the piping structure **60** according to the present embodiment, the conductance of the plurality of pipes **61** to **64** may be made the same while the passage routes and positions are made different from each other, and both the disposition with a high degree of freedom and the constant conductance may be achieved.

(41) FIG. 4 is a side view of the piping structure according to the present embodiment. As illustrated in FIG. 4, it may be seen that the pipes **61** to **64** of the piping structure **60** according to the present embodiment are configured to pass through various routes. This enables the disposition with a high degree of freedom.

(42) FIG. 5 is an exploded view illustrating one pipe of the piping structure according to the embodiment of the present disclosure in an exploded manner. In FIG. 5, the pipe **61** is given as an example, but the pipes **62** to **64** also have the same configuration.

(43) In FIG. 5, the partial pipe **611** on the process module connection side is illustrated on the left, the partial pipe **613** on the vacuum pump connection side is illustrated on the right, and the partial pipe **612** in the middle is illustrated in the center.

(44) The partial pipe **611** includes an inclined portion **611a** and a straight portion **611b**. A flange **6111** is provided at the uppermost part of the inclined portion **611a** and a flange **6112** is provided at the lowermost part of the straight portion **611b**. The flange **6111** connected to the process module **40** and the flange **6112** connected to the partial pipe **612** have different structures. In this way, as for the flanges **6111** and **6112**, an appropriate type of flange may be used according to the object to be connected.

(45) In this way, by combining the inclined portion **611a** and the straight portion **611b**, the position of the pipe may be shifted in the horizontal direction at the inclined portion **611a**. Further, by providing the straight portion **611b**, the conductance of the partial pipe **611** may be suppressed from decreasing. That is, the partial pipe **611** has a structure in which the position of the partial pipe **61** is shifted at the inclined portion **611a**, and a straight pipe is used at a portion where the position does not need to be shifted to avoid an increase in conductance.

(46) An inclination angle of the inclined portion **611a** may be set in various ways depending on the usage, and may be, for example, 20 to 70 degrees, 30 to 60 degrees, or 45 degrees with respect to the vertical direction. In FIG. 5, an example is given in which the inclined portion **611a** has an inclination angle of 45 degrees with respect to the vertical direction.

(47) The partial pipe **612** in the middle also includes an inclined portion **612a** and a straight portion **612b**, and the position of the pipe **61** is shifted at the inclined portion **612a**, and the other portions are made into straight lines which do not easily decrease the conductance.

(48) Since the upper and lower ends of the partial pipe **612** are connected to the partial pipes **611** and **613**, respectively, the same type of flanges **6121** and **6122** are used for the upper and lower ends.

- (49) The inclination angle of the inclined portion **612a** is also set to, for example, 45 degrees. The point that the inclination angle may be changed according to the usage has been described with reference to the partial pipe **611**.
- (50) The partial pipe **613** connected to the vacuum pump **50** includes a straight portion **613a**, an inclined portion **613b**, and a straight portion **613c**. The straight portion **613c** is connected to the vacuum pump **50** and configured as a straight portion to simplify the structure.
- (51) Although the inclined portion **613b** and the straight portion **613a** are different in the disposition order, the inclined portion **613b** and the straight portion **613a** have the same purpose as the inclined portions **611a** and **612a** and the straight portions **611b** and **612b** included in the partial pipes **611** and **612**.
- (52) Further, since the upper end of the partial pipe **613** is connected to the lower end of the partial pipe **612** and the lower end thereof is connected to the vacuum pump **50**, the flange **6131** provided at the upper end thereof and the flange **6132** provided at the lower end thereof have different configurations.
- (53) In this way, the pipe **61** is configured by connecting the partial pipes **611** to **613** in which the inclined portions **611a**, **612a**, and **613b** and the straight portions **611b**, **612b**, **613a**, and **613c** are combined with each other. As a result, the pipe **61** may be configured into various shapes.
- (54) Further, since this configuration is the same for the other pipes **62** to **64**, the conductance of the pipes **61** to **64** may be made the same, and various piping structures **60** may be provided according to the conditions of the clean room facility structure **30**, columns, or walls while maintaining the conductance constant.
- (55) FIG. **6** is a view illustrating an example of the configuration of the flange **6111**. The flange **6111** includes a flange portion **61111**, a center ring **61112**, and a claw clamp **61113**. The flange **6111** is configured to be able to rotate and connect the pipes due to a degree of freedom in the rotational direction with respect to the center of the pipe. In this way, it is possible to use the rotatable flange **6111** to connect the partial pipes **611** to **613** at various angles, thereby configuring the piping structure according to the present embodiment.
- (56) The structure of the flange **6132** may be the same as that of the flange **6111**, and the flanges **6112**, **6121**, **6122**, and **6131** may use rotatable flanges which are suitable for connecting pipes to each other.

Second Embodiment

- (57) FIG. **7** is a view illustrating an example of a partial pipe **61a** of a piping structure **60a** according to a second embodiment of the present disclosure. Regarding the partial pipes **611** and **612**, the pipe **61a** of the piping structure **60a** according to the second embodiment has the same structure as the partial pipe **61** according to the first embodiment. However, there is a difference in that the partial pipe **614** disposed in the pipe **61a** of the piping structure **60a** includes one straight portion **614a** and one inclined portion **614b**, whereas the partial pipe **613** disposed in the partial pipe **61** includes two straight portions and one inclined portion.
- (58) As described above, regarding all the partial pipes **611**, **612**, and **614**, a structure may be prepared in which each of the straight portions **611b**, **612b**, and **614a** and each of the inclined portions **611a**, **612a**, and **614b** are provided. Even in this case, the positions of the inclined portions **611a**, **612a**, and **614b** are shifted, and the number of the inclined portions **611a**, **612a**, and **614b** is three in total, which does not change. Thus, the position of the pipe **61a** may be changed in the same manner as the pipe **61** of the piping structure **60** according to the first embodiment. The other pipes **62** to **64** may also be the pipes **62a** to **64a** having the same configuration.
- (59) FIG. **8** is a view illustrating an example of openings **31** of a clean room facility structure **30**. The shapes of the pipes **61a** to **64a** are determined such that the pipes **61a** to **64a** may pass through the openings **31**.
- (60) FIGS. **9A** and **9B** are views illustrating a swing width in the pipe **61a**. FIG. **9A** is a plan view of the pipe **61a** as viewed from above. FIG. **9B** is a side view of the pipe **61a** as viewed from the

side.

(61) As illustrated in FIG. 9A, the lengths protruding to the sides of the inclined portions **611a**, **612a**, and **614b** determine a fluctuation range of the pipe **61a**.

(62) FIG. 9B illustrates an example of a method of setting the fluctuation range of the inclined portions **611a**, **612a**, and **614b**. In FIG. 9B, an example is given in which the length of the orthogonal projection of the inclined portion **611a** of the partial pipe **611** in the horizontal direction is set to 400 mm, the length of the orthogonal projection of the inclined portion **612a** of the partial pipe **612** in the horizontal direction is set to 650 mm, and the length of the orthogonal projection of the inclined portion **614b** of the partial pipe **614** in the horizontal direction is set to 650 mm. The fluctuation range of the inclined portion is exactly the length of the orthogonal projection in the horizontal direction, but hereinafter, may be abbreviated as a “fluctuation range.”

(63) The uppermost partial pipe **611a** of the pipe **61a** is a portion through which the opening **31** of the clean room facility structure **30** illustrated in FIG. 8 is passed, and the fluctuation range is also limited by the size of the opening **31**. Therefore, the fluctuation range is set as small as 400 mm.

(64) Meanwhile, in the partial pipes **612** and **614**, since there is no such restriction on the fluctuation range, a large fluctuation range of 650 mm is set.

(65) With such a setting, in the partial pipe **611**, when it is assumed that the opening **31** of the clean room facility structure **30** is simply passed through and the fluctuation width is zero, the route of the pipe **61a** may be freely set within the range of $650\text{ mm} \times 2 = 1300\text{ mm}$.

(66) In this way, when there are restrictions due to the environment of the room, it is possible to set the restricted portion to a small fluctuation range and the unrestricted portion to a large fluctuation range to secure a sufficient fluctuation range as a whole.

(67) Further, when the inclination angles of the inclined portions **611a**, **612a**, and **614b** are set to be constant, since the fluctuation range is determined by the lengths of the inclined portions **611a**, **612a**, and **614b**, the lengths of the inclined portions **612a** and **614b** of the partial pipes **612** and **614** are set to be longer than that of the inclined portion **611a** of the partial pipe **611**.

(68) In this way, the fluctuation range of the partial pipe may be variously set according to the usage. For example, as illustrated in FIGS. 9A and 9B, when there is a restriction on the opening **31** of the clean room facility structure **30**, it is possible to reduce the fluctuation range of the inclined portion **611a** of the uppermost partial pipe **611** to pass, equally increase the fluctuation ranges of the inclined portion **612a** of the second partial pipe **612** and the inclined portion **614b** of the third partial pipe **614**, and making the shapes of the entire partial pipes **612** and **614** the same, thereby reducing the processing cost. In FIGS. 9A and 9B, such a configuration is illustrated.

(69) When is no such restriction, it is possible to set the fluctuation ranges of the inclined portions **611a**, **612a**, and **614b** of the three partial pipes **611**, **612**, and **614** to be the same. In this way, the fluctuation ranges may freely set according to the usage.

(70) FIG. 10 is a view illustrating a concept of route setting. As illustrated in FIG. 10, it is possible to follow an arbitrary route within the circle of the total fluctuation range of the partial pipes **612** and **614** after passing through the openings **31** of the clean room facility structure **30**. Therefore, the pipes **61a** to **64a** may be installed to have the highest space efficiency within the circle.

(71) FIGS. 11A and 11B are views illustrating an example of the shape of a pipe **61a** in which a fluctuation range is set to be small. As illustrated in FIGS. 11A and 11B, the pipe **61a** is shaped such that the portion moved by the partial pipe **612** is returned by the partial pipe **614**. When the pipe **61a** is set to such a shape, the piping structure **60a** having a small fluctuation range may be obtained.

(72) FIGS. 12A and 12B are views illustrating an example of the shape of a pipe **61a** in which a fluctuation range is set to be large. As illustrated in FIGS. 12A and 12B, the pipe **61a** is shaped to move to the right in the partial pipe **612** and then further to the right in the partial pipe **614**. When the pipe **61a** is set to such a shape, the piping structure **60a** having a large fluctuation range may be obtained.

(73) In this way, the fluctuation range of the pipe **61a** may be set variously depending on how the partial pipes **611**, **612**, and **614** are connected, and the conductance of each pipe **61a** to **64a** of the piping structure **60a** may be set to be constant while dealing with various restrictions.

(74) Further, in the present embodiment, an example of three partial pipes has been described, but two partial pipes may be sufficient depending on the required layout. It is also possible to have four partial pipes. However, considering the ease of connection, the degree of freedom of structure, and the degree of decrease in conductance, it may be said that it is often desirable to have three partial pipes.

(75) Further, the processing apparatus according to the embodiment of the present disclosure may be applied to various process modules for processing while performing a vacuum exhaust, and may be applied to various film forming apparatuses, annealing apparatus, and etching apparatuses.

(76) According to the present disclosure, the conductance of a plurality of pipes may be made the same.

(77) From the foregoing, it will be appreciated that various embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A piping structure comprising: a plurality of pipes configured to connect a plurality of process modules disposed adjacent to each other at a first room and a plurality of vacuum pumps disposed at a second room below the first room to be corresponding to the plurality of process modules, respectively, wherein the plurality of pipes are divided into a plurality of regions in a height direction, and the plurality of pipes used for regions in an identical height have an identical shape, each of the plurality of pipes includes partial pipes provided for each of the plurality of regions in the height direction, respectively, and the partial pipes are connected to each other to constitute each of the plurality of pipes, each of the partial pipes includes an inclined portion provided to have an inclination and a straight portion extending in a vertical direction, the straight portions of the plurality of partial pipes are parallel to each other, the plurality of pipes pass through holes defined in a structure spaced apart from the first room and the second room, respectively, and in each of the plurality of pipes, a first inclined portion of a first partial pipe and a second inclined portion of a second partial pipe at opposite ends of a first straight portion of the first partial pipe disposed in the hole of the structure have different lengths from each other when viewed from above.
2. The piping structure according to claim 1, wherein a direction of the inclined portion is different for each of the plurality of pipes.
3. The piping structure according to claim 2, wherein each of the plurality of pipes includes three partial pipes.
4. The piping structure according to claim 3, wherein in the three partial pipes, lengths of inclined portions of second and third partial pipes are larger than a length of an inclined portion of an uppermost partial pipe.
5. The piping structure according to claim 4, wherein the second and third partial pipes have an identical shape.
6. A processing apparatus comprising: a plurality of process modules disposed adjacent to each other at a first room; a plurality of vacuum pumps disposed in a second room below the first room and provided to be corresponding to the plurality of process modules; and a plurality of pipes configured to connect the plurality of process modules and the plurality of vacuum pumps, respectively, wherein the plurality of pipes are divided into a plurality of regions in a height direction, and the plurality of pipes used for regions in an identical height have an identical shape,

each of the plurality of pipes includes partial pipes provided for each of the plurality of regions in the height direction, respectively, and the partial pipes are connected to each other to constitute each of the plurality of pipes, each of the partial pipes includes an inclined portion provided to have an inclination and a straight portion extending in a vertical direction, the straight portions of the plurality of partial pipes are parallel to each other, the plurality of pipes pass through holes defined in a structure spaced apart from the first room and the second room, respectively, and in each of the plurality of pipes, a first inclined portion of a first partial pipe and a second inclined portion of a second partial pipe at opposite ends of a first straight portion of the first partial pipe disposed in the hole of the structure have different lengths from each other when viewed from above.

7. The processing apparatus according to claim 6, wherein a direction of the inclined portion is different for each of the plurality of pipes.

8. The processing apparatus according to claim 7, wherein each of the plurality of pipes includes three partial pipes.

9. The processing apparatus according to claim 8, wherein in the three partial pipes, lengths of inclined portions of second and third partial pipes are larger than a length of an inclined portion of an uppermost partial pipe.

10. The processing apparatus according to claim 9, wherein the second and third partial pipes have an identical shape.

11. The processing apparatus according to claim 6, wherein each of the plurality of process modules is configured to process a substrate, and the structure between the first room and the second room is a clean room facility structure.
